

Power for Chi-Squared Tests

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Introduction

Power analysis for chi-squared tests — goodness-of-fit against a hypothesised distribution and independence in a two-way contingency table — uses the non-central chi-squared distribution under the alternative hypothesis. The standardised effect size for these tests is Cohen's w , which combines the difference between hypothesised and expected cell proportions into a single number that maps directly to the non-centrality parameter of the test statistic. Power calculations are essential in the design of survey research, epidemiological category-association studies, market-segmentation analyses, and any setting where the central question is whether categorical proportions differ from a reference or vary across rows of a contingency table.

Prerequisites

A working understanding of chi-squared goodness-of-fit and contingency-table tests, the chi-squared distribution and its non-central generalisation, and Cohen's w as the standardised effect-size measure.

Theory

Cohen's w is

$$w = \sqrt{\sum_i \frac{(p_{1i} - p_{0i})^2}{p_{0i}}},$$

where p_{0i} are the cell proportions expected under the null hypothesis and p_{1i} under the alternative. Equivalently, $w = \sqrt{\chi^2/n}$ at the expected effect. Under the alternative, the test statistic follows a non-central chi-squared with the test's standard degrees of freedom and non-centrality parameter $\lambda = nw^2$. Conventional benchmarks for w are 0.10 (small), 0.30 (medium), and 0.50 (large).

Assumptions

Observations are independent, expected cell counts are large enough that the chi-squared approximation is valid (typically ≥ 5 per cell, though stricter thresholds apply for small tables), and the cell proportions under the alternative hypothesis are pre-specified.

R Implementation

```
library(pwr)

pwr.chisq.test(w = 0.3, df = 3, sig.level = 0.05, power = 0.80)

pwr.chisq.test(w = 0.3, df = 4, sig.level = 0.05, power = 0.80)

p0 <- c(0.25, 0.25, 0.25, 0.25)
p1 <- c(0.40, 0.30, 0.20, 0.10)
w <- sqrt(sum((p1 - p0)^2 / p0))
```

```
w  
pwr.chisq.test(w = w, df = 3, power = 0.80)
```

Output & Results

`pwr.chisq.test()` returns the required sample size to achieve target power against a specified w , α , and df. For a medium effect $w = 0.3$ with 3 df (4-category goodness-of-fit), $n = 122$; with 4 df (a 3×3 contingency table), $n = 133$. Computing w directly from hypothesised cell proportions is the recommended workflow because it ties the calculation to substantive distributional assumptions.

Interpretation

A reporting sentence: “To detect a deviation from equal proportions across four categories of medium magnitude (Cohen’s $w = 0.30$, equivalent to expected proportions (0.40, 0.30, 0.20, 0.10) vs. uniform null) at 80 % power and $\alpha = 0.05$, $n = 122$ observations are required. With 5 % attrition, the protocol enrolls 130 participants. Sensitivity analyses across $w \in [0.20, 0.40]$ are reported in the supplement to bracket uncertainty about the true effect size.” Always state both w and the underlying proportional hypothesis.

Practical Tips

- Convert real-world quantities — relative risks, odds ratios, differences in proportions — to Cohen’s w via the contingency-table formula rather than guessing; using a generic “medium” benchmark when the underlying scientific question implies a different magnitude is a recurring source of mis-powered studies.
- Required sample size grows roughly linearly with degrees of freedom at fixed w ; spreading effects across more cells (e.g., 5×5 vs 2×2) reduces the per-cell signal and inflates the sample-size requirement.
- For small expected cell counts that would invalidate the chi-squared approximation, plan to use Fisher’s exact test and estimate power by Monte Carlo simulation; the asymptotic chi-squared power calculation is unreliable in the small-sample regime where exact tests are needed.
- If the alternative-hypothesis cell proportions are themselves uncertain, pre-specify a sensitivity range and report power across the range; this is increasingly expected by reviewers and protects against under-powering when the assumed proportions turn out to be optimistic.
- Chi-squared power calculations are approximate (the non-central chi-squared assumes large samples); simulation-based power is safer for complex designs, sparse-data scenarios, or when the test statistic deviates from the standard Pearson form.
- For one-way exact-binomial tests of a single proportion, use specialised tools like `pwr.p.test()` or `binom::power.binom()` rather than the chi-squared approximation; the exact methods are more accurate at small samples.

R Packages Used

`pwr::pwr.chisq.test()` for the canonical Cohen’s w chi-squared power calculation; `WebPower::wp.chisq()` for an alternative interface; `MESS::power_prop_test()` for two-proportion specific power; `binom` for exact-binomial sample-size tools; `Mediana` for trial-design simulation including chi-squared-test power across more complex designs.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation